



Dynamic service migration and resource management for vehicular clouds

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Abstract

Vehicular cloud computing has received huge attention in business and scientific communities, which integrates two emerging fields, namely cloud computing and vehicular ad hoc networks. It acts as a data center by using the underutilized resources of the networked vehicles. Moreover, many studies suggest these vehicles as potential candidates for hosting virtual machines (VMs). As a result, a VM can be set up using the vehicular resources, and it can be transferred from one vehicle to another vehicle to continue its execution, under some circumstances. It enables the vehicular environment to provide services to the user requests, that are submitted to the cloud. However, the mapping of such requests to the VMs (or hosted vehicles) and service migration is very much challenging, and not well-studied in the literature. In this paper, we propose a dynamic service migration algorithm for vehicular clouds. The algorithm consists of three phases, estimation, assignment and migration. The performance is carried out through simulation runs using two scenarios of six datasets, and compared with three well-known algorithms, namely vehicular VM migration-uniform, round robin and mobility and destination workload aware migration using four performance measures. The comparison results followed by statistical validation using *T* test show the superiority of the proposed algorithm over the existing algorithms.

Keywords Cloud computing · Vehicular ad hoc network · Vehicular cloud · Migration · Resource management · Roadside unit

1 Introduction

The emergence of cloud computing led to the provisioning of the computing facilities and resources as services to the users in an on-demand and pay-per-use basis (Buyya et al. 2009; Panda and Jana 2015; Pande et al. 2016; Wei

et al. 2010a, 2014, 2010b). These services are rendered in the form of virtual machines (VMs) that become a part of physical resources (or data centers), which increases the scalability. Alternatively, a user can set up the VM environment, without concerning the underlying cloud infrastructure (Refaat et al. 2016; Panda and Jana 2018; Panda et al. 2018; Rahimi et al. 2014; Mastelic et al. 2015). The cloud computing services have drastically increased and covered a wide range of possibilities with the integration of devices via the Internet. Here, the devices are not limited to tablets, personal computers, laptops and personal digital assistants, they also include smart vehicles, smart gadgets and many more. As a result, cloud computing is more appealing in business and scientific communities (Panda et al. 2019; Panda and Jana 2019a, b).

In recent years, vehicular networks have gained immense popularity based on the increasing involvement of smart vehicles. A smart vehicle is equipped with an on-board unit, which consists of processing unit, storage unit, global positioning system, radar device, sensors, communication devices, digital map, cameras and many more (Kim et al.

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